Terrain, Not Latitude: Sunlight and Earth Contact at South-Polar Landing Sites, With the Uncertainty Each Number Carries

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POLAR WINDOW
NASA Space Apps Challenge 2026, CLPS Lunar Mission Browser

***Abstract*—Commercial Lunar Payload Services (CLPS) deliveries are converging on the lunar south pole, where the Sun circles within a few degrees of the horizon and terrain, not latitude, decides whether a lander has power and a line of sight to Earth. We describe the geometry engine of POLAR WINDOW, a CLPS mission browser that reports four numbers for any site and window: hours of sunlight, hours of Earth contact, the longest unbroken dark gap and the longest unbroken radio blackout, never combined into a score. Each skyline is ray-marched once from Lunar Orbiter Laser Altimeter terrain; everything after is a lookup and one vectorised SPICE call per body. The contribution is the uncertainty model: per-bearing elevation error, a near-field noise floor, grazing crossings reported as untimeable, and window-censored gaps reported as lower bounds. We also measure what the tolerance omits: a tuning constant moves a summit's sunlight from 574.399 h to 482.33 h.**

I. Problem

At Connecting Ridge (−89.438° latitude) the Sun never rises: across the 708.75 h window from 2027-03-14 used throughout, its elevation stays between −1.607° and −0.703 1° (Figure 1). A sun calculator with a level horizon reports no sunlight. Yet the site is sunlit for 81.04 % of the window, because it stands 1 957.5 m above the 1 737.4 km reference sphere and looks *down* at its skyline, whose mean over 1 440 bearings is −2.368 1°. At the pole the horizon decides [1]–[4].

The second trap is the coordinate. A crater's published coordinate is its centre, which is its floor: the 80 m LOLA model returns −2 767.3 m, −3 315.9 m and −3 931.7 m at the anchors for Shackleton, Haworth and Shoemaker. The 15 candidate sites are therefore resolved from terrain: the most open-skied post, sloped at most 12°, in the top elevation decile around each anchor. Eight of the 20 records in the project's CLPS dataset target the south pole.

II. Method

a) Horizon mask: Algorithm 1 marches 1 440 rays at 0.25° to 50 km, 555 samples each, starting two posts out, beyond the site's own pixel. Each elevation angle subtracts the curvature drop,

$$\theta_k = \arctan \frac{h(d_k) - h_0 - d_k^2/2R}{d_k}, \qquad R = 1\,737.4\,\text{km}, \quad (1)$$

which is 115.11 m at 20 km, and is never clamped at zero (Figure 2). Terrain is LDEM_80S_20M within 20 km and LDEM_80S_80M beyond [5], [6].

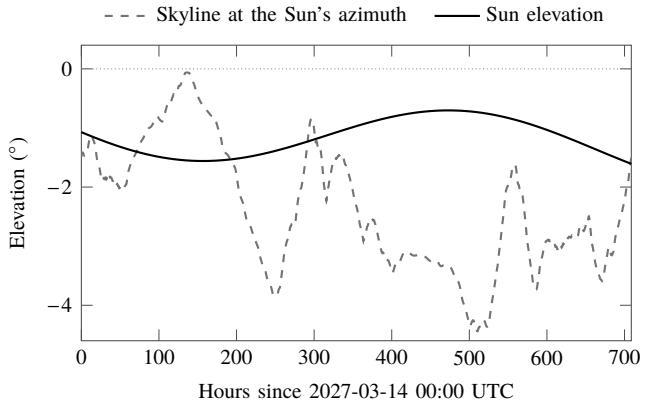


Fig. 1. At Connecting Ridge the Sun stays below level (dotted) for the whole window, yet it clears the site's own skyline for 81.04 % of it, because that skyline lies below level on every bearing. Data: `sun_vs_skyline.dat`, 2 836 samples at 900 s.

b) Visibility: One SPICE [7], [8] `spkpos` call per body, through SpiceyPy [9], gives Sun and Earth from DE440 [10] in MOON_ME [11], projected onto the site's east, north and up unit vectors, which stay exact at the pole. The margin $m = e - H(a)$ subtracts the mask at the body's azimuth from its elevation. Discs are finite: the angular radius ρ spans 0.266 0° to 0.268 3° for the Sun and 0.902 5° to 0.996 3° for Earth, so a sample is lit if $m > \rho$, dark if $m < -\rho$ and otherwise penumbral (Figure 3), as a median 8.44 % of samples are.

c) Intervals and metrics: Sign changes of m are bracketed at 900 s and bisected 40 times; half-open runs partition the window. The four numbers are two totals and two longest runs, never combined.

d) Why precompute: The mask depends only on the site and costs a median 128.653 ms; a cold `/metrics` request costs 10.162 ms (Table I; Figure 4). The mask is built once per site and engine version, and a request is a lookup plus one ephemeris call.

III. Uncertainty

a) Per-tier error: Each bearing carries the vertical error σ of the tier that set its skyline, 2 m on the 20 m tier or 4 m on the 80 m tier, as $\delta(a) = \arctan(\sigma/r^*)$. At Connecting Ridge

Algorithm 1: Horizon mask for one site

Input: height sampler $h(x, y)$, vertical-error sampler $\sigma(x, y)$, site height h_0, post spacing p
Output: per bearing a: skyline $\theta^*(a)$, range $r^*(a)$, tolerance $\delta(a)$

$d_1 \leftarrow 2p$; $d_{k+1} \leftarrow d_k + \max(5\,\text{m},\ 0.01\,d_k)$ while $d_{k+1} \le 50\,\text{km}$
for $a \in \{0, 0.25°, \ldots, 359.75°\}$ **do**
  $(x_k, y_k) \leftarrow d_k(\sin a, \cos a)$ for all k
  if *no* $h(x_k, y_k)$ *is defined* **then** $\theta^*(a) \leftarrow$ smooth-sphere dip; **continue**
  $\theta_k \leftarrow$ Equation (1); $\phi_k \leftarrow \arctan(\sigma_k / d_k)$
  $E \leftarrow \{k : \theta_k > \phi_k \ \vee \ \phi_k < 0.25°\}$
  $k^* \leftarrow \arg\max_{k \in E} \theta_k$
  $\theta^*(a) \leftarrow \theta_{k^*}$; $r^*(a) \leftarrow d_{k^*}$; $\delta(a) \leftarrow \arctan(\sigma_{k^*} / d_{k^*})$

Figure pending: `figures/raymarch-geometry.pdf`

Fig. 2. Each sample's elevation angle is measured after the surface has curved away by $d^2/2R$, so a site above its surroundings has a skyline below level, and the mask keeps that negative angle.

746 of 1 440 bearings are set on the coarse tier, so the site reports 4 m, not the finest tier's figure.

b) Near-field noise floor: A 2 m error at 40 m is 2.862 4° of apparent skyline (Figure 5). A sample may set the skyline only if it clears that floor, or if the floor is below 0.25°, which holds beyond 458.4 m for 2 m.

c) Grazing crossings: An edge's timing tolerance is $\delta t = \delta(a)/|\mathrm{d}m/\mathrm{d}t|$ (Figure 6). Below a crossing rate of 0.002 5 °/h, or with δt above 24 h, the crossing time is not knowable from these data, and the edge is reported as *grazing*, without a tolerance, rather than with a large one. Faustini rim's longest dark gap and blackout both graze (Table II; Figure 7).

d) Censoring: A run touching the window edge was ended by the window, so its duration is a lower bound. Five of the 15 sites have a censored longest night. A bound claims $[v, \infty)$; an estimate claims $[v - u, v + u]$. A site beats another only when its interval clears the other's, so on longest night Connecting Ridge ((121.57 ± 17.09) h) ranks 1–2, de Gerlache rim ($\geqslant$ 112.17 h) 1–3 and Malapert massif ($\geqslant$ 287.76 h) 2–3. Reading the bound as an estimate would make de Gerlache rim the best site and size its battery for a night shorter than the real one.

Figure pending: `figures/state-decision.pdf`

Fig. 3. A sample is lit or dark only when the whole disc is clear of, or hidden by, the skyline; the band in between is reported as penumbral, not rounded to either side.

TABLE I
PRECOMPUTE VERSUS REQUEST COST, CONNECTING RIDGE, REFERENCE WINDOW AT 900 s. APPLE M3 (MAC15,12), MACOS 27.0, PYTHON 3.12.14. VALUES AS WRITTEN TO `data/timing.csv`.

Stage	Median (ms)	p95 (ms)	N
Once per site (worker)			
Ray-march, 1 440 bearings	128.653	130.154	75
Load 20 m window	22.223	25.435	45
Load 80 m product	399.12	420.001	3
Fetch 20 m window from PDS	not measured		
Per request (API)			
Ephemeris, two bodies	6.552	6.704	50
Sun track and crossings	1.938	2.067	50
Earth track and crossings	0.694	0.707	50
Four numbers and autonomy	0.008	0.010	50
HTTP `/metrics`, cache miss	10.162	11.249	30
HTTP `/metrics`, cache hit	0.782	0.884	30
HTTP `/timeline`, cache miss	14.369	14.930	30

e) What the tolerance omits: On 424 of Connecting Ridge's bearings the ground falls away from the summit, so the skyline is set at 460 m, the first sample the floor trusts: the rule, not a ridge, fixes it. Both edges of the longest night lie there, hence ±17.09 h. Moving the threshold to 0.125° or 0.5° changes sunlight to 611.823 h or 482.33 h (Table III), far outside that tolerance.

IV. Validation

a) Golden fixtures: Three sites chosen for different physics are rebuilt from terrain on disk, not compared with a stored mask, to 0.05° on the skyline, 72 s on arc edges and the larger of 360 s and 0.1 % on the four numbers. Connecting Ridge's rebuilt mask is identical to the stored one.

b) Properties: Interval code knows no astronomy, so Hypothesis [12] draws up to 300 boolean series per property and checks partition, alternation, contiguity, inversion symmetry and monotonicity, with degenerate shapes pinned.

c) Adapter: The browser's adapter tests assert its hours against the seconds the backend golden pins.

d) Engine stamp: The engine version, hz-2/vis-2, is in every cache key, stored mask and response, so geometry from another engine is a readable 409, not a stale answer. Table IV

Figure pending: `figures/timing-breakdown.pdf`

Fig. 4. The mask costs 12.66 times a complete cold request and depends only on the site, so it is built once; every request is a lookup and an ephemeris call.

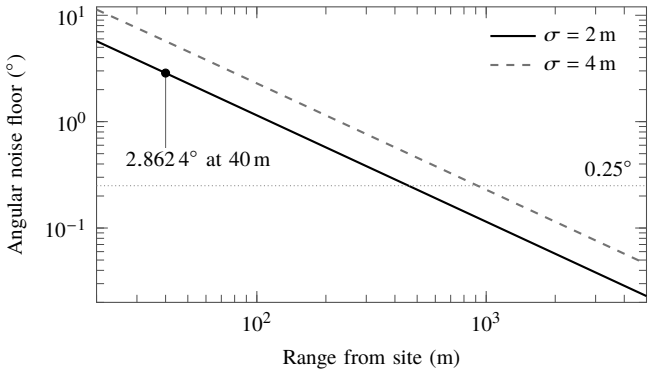


Fig. 5. Close to the site, elevation error alone produces skylines of several degrees. Samples are trusted unconditionally only where the floor $\arctan(\sigma/r)$ falls below 0.25° (dotted): beyond 458.4 m for 2 m and 916.7 m for 4 m. Data: `noise_floor.dat`.

lists the results; the one failure is the OpenAPI snapshot, stale against uncommitted schema changes.

V. Limitations

The 120 h survival limit is a browser placeholder for a thermal model the system does not have; a per-lander thermal and battery model would replace it. Elevation error is constant per tier, not per pixel; the LOLA count products (LDEC_80S_20M and LDEC_80S_80M [5]) would localise it. At summits the near-field threshold outweighs the reported tolerance; a lander-height term and 5 m site maps [13] would bound it. Illumination and shadow tiles use a five-ring horizon proxy and are labelled so; per-pixel masks on the 80 m grid mean 57 760 000 ray-marches, 86.0 d single-threaded. No screen-reader user has tested the arc announcements; only such users can. Fetching a 20 m window from PDS was not timed; timing `make seed` on an empty volume would.

VI. Conclusion

POLAR WINDOW answers a CLPS planner with four numbers per site and window, each carrying a tolerance, the word *grazing*, or a lower bound, and none folded into a score. Geometry costs 128.653 ms once per site; a cold request then takes 10.162 ms. The uncertainty model separates crossings the data can time from those they cannot, and bounds from estimates, which changes how sites rank. Its own sensitivity test names the next step: at summits the near-field rule moves sunlight by more than the reported tolerance, and that must be bounded before these numbers choose a landing site.

Figure pending:
`figures/horizon-polar-connecting-ridge.pdf`

Fig. 6. Connecting Ridge's skyline lies below level on every bearing. The crossings that bound its longest night fall where the skyline tolerance is largest, so they carry the widest timing bands.

TABLE II
The four numbers for three contrasting sites over the reference window, in hours, rounded to two decimals from `data/sites_four_numbers.csv`. ⩾ marks a lower bound; *grazing* marks an edge that cannot be timed.

Metric (h)	Connecting Ridge	Malapert massif	Faustini rim
Sunlight	574.40 ± 17.09	394.16 ± 7.25	103.07 ± 24.00
Earth contact	430.28 ± 5.26	570.61 ± 7.63	278.57 ± 33.94
Longest dark	121.57 ± 17.09	⩾ 287.76	605.68, grazing
Longest blackout	278.47 ± 5.26	138.14 ± 7.63	430.18, grazing

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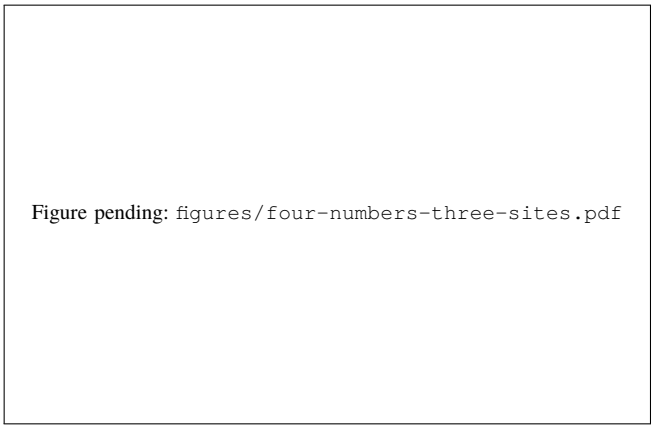


Fig. 7. A measured value carries an error bar, a censored value is drawn as a bound with an open end, and a grazing value has no bar at all: three different claims that a single point would hide.

TABLE III
CONNECTING RIDGE UNDER THREE VALUES OF THE NEAR-FIELD THRESHOLD, AS WRITTEN TO `data/sensitivity.json`. THE ENGINE USES 0.25°.

Threshold (°)	Trusted from (m)	Mean skyline (°)	Sunlight (h)	Longest dark (h)
0.125	916.7	−2.656 6	611.823	84.145
0.25	458.4	−2.368 1	574.399	121.569
0.5	229.2	−1.775 6	482.33	206.515

TABLE IV
WHAT `make test`, `make test-fe` AND `make lint` PRINTED ON 2026-09-24, WITH THE BACKEND SUITES ALSO RUN PER DIRECTORY.

Command	Suite	Passed	Failed
`make test`	unit	159	0
	integration	56	1
	golden	16	0
	total, 6.41 s	231	1
`make test-fe`	adapter, keys, exports	41	0
`make lint`	ruff: no findings, 80 files formatted		
	mypy strict: no issues in 80 files		
	import contracts: 8 kept, 0 broken		

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